Global Currency Power Index (CCPI) – A Multi-Dimensional Economic Analysis

Data Analytics & Visualisation Data9005

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Date: 06-03-2026

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1. Introduction

The strength of a country's currency shows how well its economy is doing compared to countries. You cannot just look at one thing like how money the country makes the exchange rate or inflation to get the full picture. This report looks at how the economy's doing all around the world. It tries to create a way to measure how strong a country's currency is. This is called the **Composite Currency Power Index**.

The main question is: which countries have gotten stronger or weaker from 2000 to 2024? What do we see when we group countries by how their economies work?

Dataset: World Bank – World Development Indicators (*World Development Indicators | DataBank*, 2024).

DataSetURL: https://databank.worldbank.org/source/world-development-indicators

Variables selected: GDP (USD), Inflation (CPI %), Official Exchange Rate (LCU/USD), Trade as % of GDP, Total Reserves (incl. Gold), Current Account Balance (% of GDP).

2. Literature Review and Prior Visualisations

World Bank WDI dataset is one of the widely used economic datasets globally.
The Gapminder (*Gapminder Tools*, no date) has produced bubble charts of GDP per capita versus life expectancy using WDI data.
Our World in Data regularly publishes choropleth maps and time-series charts of WDI variables.
Hans Roslings famous TED presentations used animated scatter plots of WDI data.

However, a composite structural currency power index (CCPI) combining GDP, reserves, inflation, trade openness, and current account balance into a single weighted, normalised metric and then visualising it dynamically across time, through PCA clustering, animated bar races, and repositioning scatter plots, represents an original contribution not found in existing published visualisations for this dataset.

The animated 'bar race' chart has appeared in general economic journalism (e.g., GDP race charts), but the application here to a bespoke composite index with multi-year animation across 100+ countries is novel in this context.

3. CCPI Construction

The CCPI is inspired by the IMF's composite weighting methodology (Mialou, no date). The construction pipeline:

- **Data Cleaning:** ".." placeholders replaced with NaN; countries with <15 valid observations excluded.
- **Forward Fill:** Missing values within a country's time series filled forward — appropriate for persistent macroeconomic variables.
- **Min-Max Normalisation:** All six variables scaled to [0,1] across all countries and years simultaneously for cross-country comparability.
- **Inflation :** The Inflation dimension is inverted (1. Scaled value) so that lower inflation contributes positively to the index.
- **Weighted Sum :** A weighted sum is computed , GDP (30%), Reserves (25%),Trade % GDP (15%), Current Account (15%), Inflation (15%).

Key construction code:

```
# 4. Construct CCPI index

features = ["GDP", "Reserves", "TradePercentGDP",
            "CurrentAccount", "Inflation"]

scaler = MinMaxScaler()
df_scaled = df.copy()
df_scaled[features] = scaler.fit_transform(df[features])

# Convert inflation: lower inflation = better score
df_scaled["Inflation"] = 1 - df_scaled["Inflation"]

# Composite index (weights)
df_scaled["CCPI"] = (
    df_scaled["GDP"] * 0.30 +
    df_scaled["Reserves"] * 0.25 +
    df_scaled["TradePercentGDP"] * 0.15 +
    df_scaled["CurrentAccount"] * 0.15 +
    df_scaled["Inflation"] * 0.15
)

df_scaled[["Country", "Year", "CCPI"]].head()
```

Table1: Key construction Code snippet of CCPI

Min-Max was chosen over Z-score standardisation because it produces bounded [0,1] scores, keeping the composite directly interpretable. Z-score is sensitive to extreme outliers (USA, China on GDP/Reserves), which would compress variation for all other countries.

4. Visualisations and Commentary

4.1 Correlation Heatmap

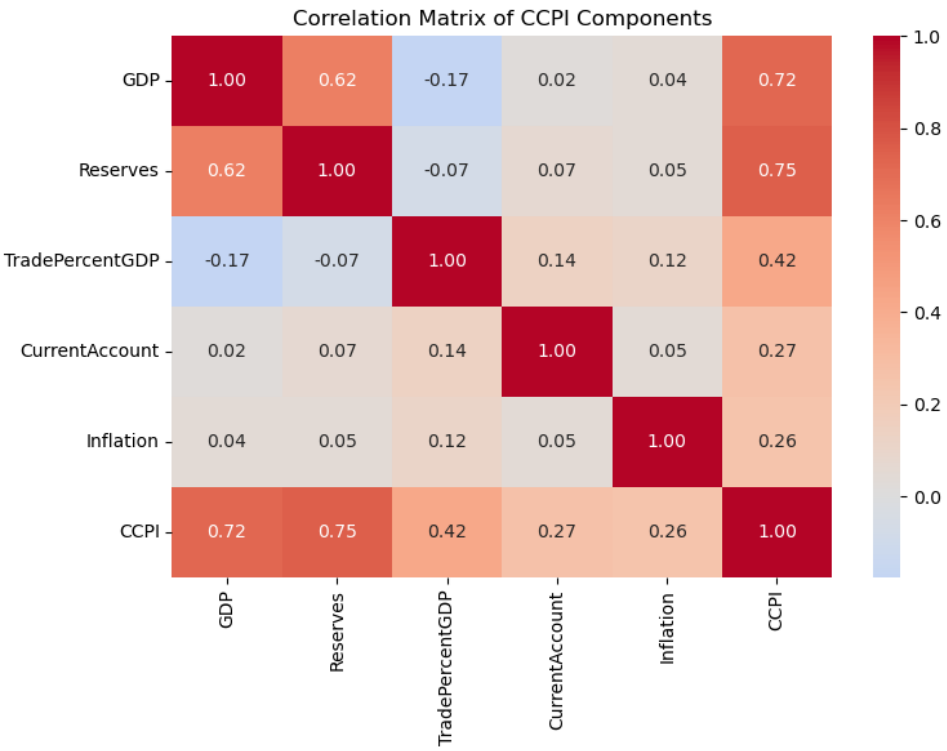

Figure 1: Correlation Matrix of CCPI Components — sns.heatmap(), coolwarm palette

GDP and Reserves are the strongest correlated pair (r = 0.62), expected since large economies accumulate proportionally larger reserve buffers. Both drive the CCPI most strongly (r = 0.72 and 0.75 respectively). Inflation shows near-zero correlations with GDP (r = 0.04) and Reserves (r = 0.05), statistically justifying all six variables as independent dimensions rather than redundant signals.

4.2 CCPI Trajectory Lines — Top 5 Economies

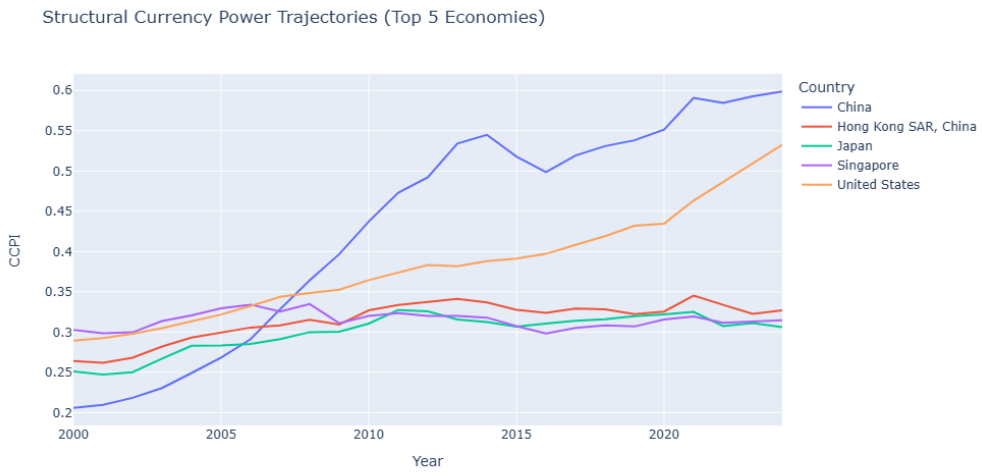

Figure 2: Structural Currency Power Trajectories — Top 5 Economies (2000–2024)

China's growth is really standing out. In 2000 it was not among the countries but by 2008 it had surpassed the United States. China kept its lead through 2024.This was because of its growing economy, savings, which peaked at over $4 trillion and steady trade surpluses. The United States stayed in a position. The 2008 financial crisis was a turning point, for all economies.

4.3 Animated Choropleth Map

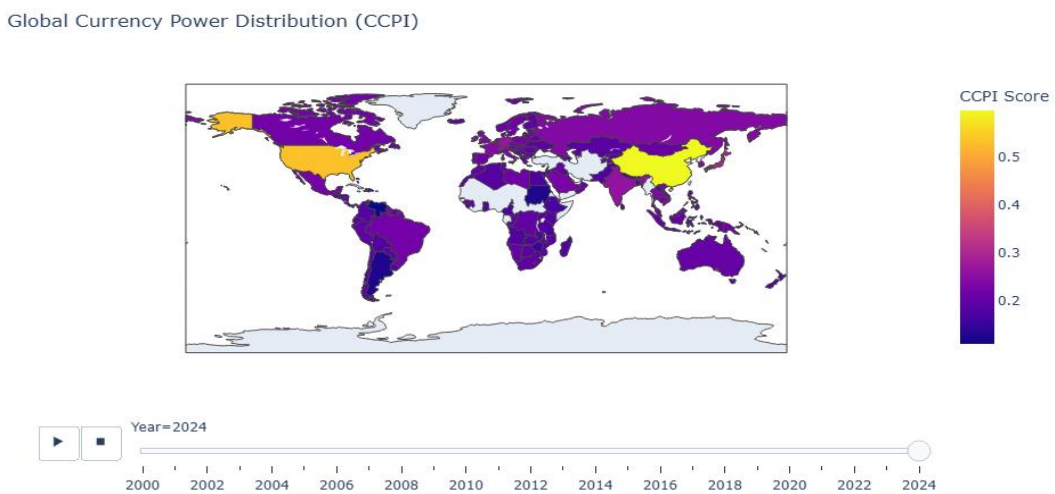

Figure 3: Animated Choropleth Map — Global CCPI Distribution (2000–2024), Plasma scale

East Asian economies and the United States consistently occupy the highest CCPI quintile. Sub-Saharan Africa and parts of South Asia remain persistently in the lowest quintile throughout the 25-year period, underscoring how structural currency power is highly path-dependent. The Plasma scale was chosen for perceptual uniformity and accessibility to colour vision deficiencies.

4.4 Animated Bubble Chart — Structural Landscape

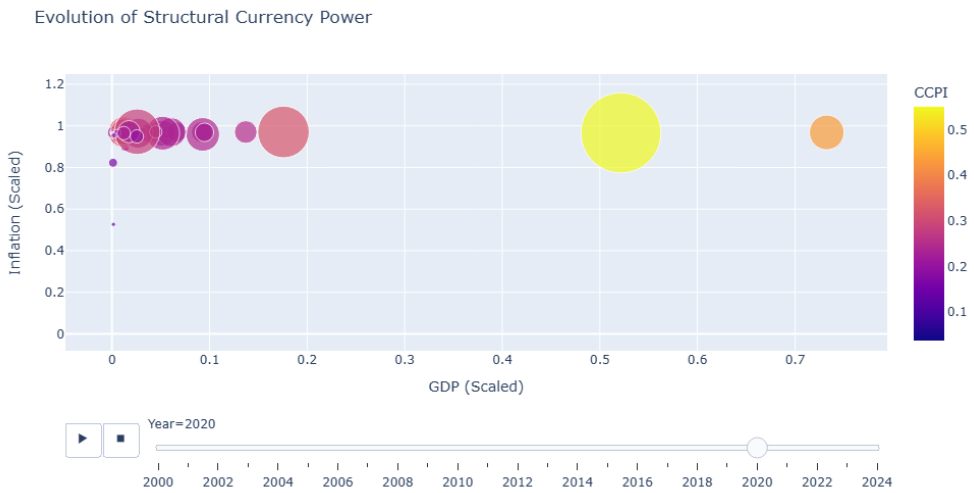

Figure 4: Animated Bubble Chart — GDP vs Inflation (bubble = Reserves, colour = CCPI)

Four dimensions are encoded simultaneously: GDP (x), monetary stability (y, inverted inflation), reserves (bubble size), and composite power (colour). The upper-right quadrant represents ideal currency strength. China's bubble moves decisively into this quadrant from 2005 onward. The 2008 GFC is visible as a temporary downward shift for several economies. Outliers — resource-dependent states with reserves disproportionate to GDP — are easily identifiable.

4.5 PCA + K-Means Clustering — Complex Implementation

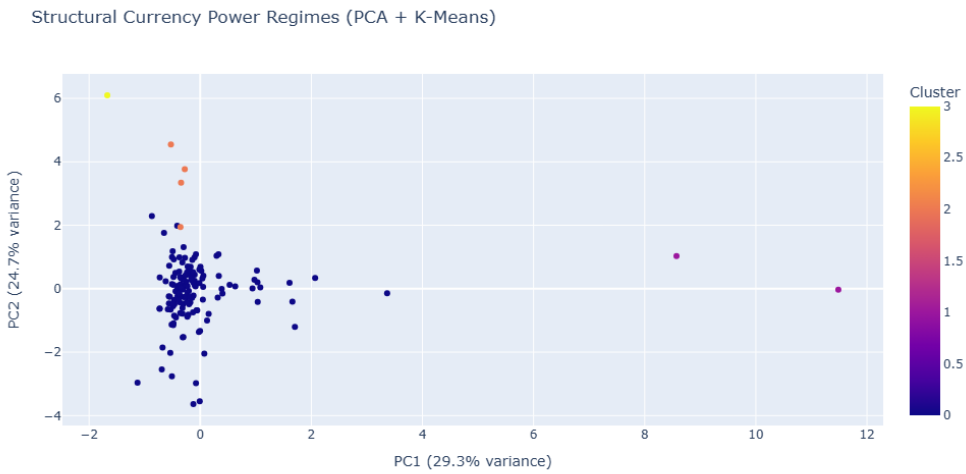

Figure 5: PCA + K-Means Clustering — Structural Currency Power Regimes

This is the most technically complex visualisation. The pipeline chains StandardScaler normalisation, K-Means clustering (K=4, selected via elbow curve), and PCA dimensionality reduction into a single interpretive scatter plot (Jolliffe, 2002f, 2002b, 2002a, 2002d, 2002e, 2002c).

```
# 9. Clustering structural regimes

latest_year = df_scaled["Year"].max()
df_cluster = df_scaled[df_scaled["Year"] == latest_year].copy()

cluster_features = ["GDP", "Inflation", "ExchangeRate",
                    "TradePercentGDP", "Reserves", "CurrentAccount"]

X = df_cluster[cluster_features]

# Standardize
scaler_std = StandardScaler()
X_scaled = scaler_std.fit_transform(X)

# K-means clustering
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
df_cluster["Cluster"] = kmeans.fit_predict(X_scaled)

# PCA for 2D visualization
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
df_cluster["PC1"] = X_pca[:, 0]
df_cluster["PC2"] = X_pca[:, 1]
```

```python
fig = px.scatter(
    df_cluster,
    x="PC1",
    y="PC2",
    color="Cluster",
    hover_name="Country",
    title="Structural Currency Power Regimes (PCA + K-Means)"
)
fig.update_layout(
    xaxis_title=f"PC1 ({pca.explained_variance_ratio_[0]*100:.1f}% variance)",
    yaxis_title=f"PC2 ({pca.explained_variance_ratio_[1]*100:.1f}% variance)"
)
fig.show()

# PCA loadings for interpretation
loadings = pd.DataFrame(
    pca.components_.T,
    columns=["PC1", "PC2"],
    index=cluster_features
)
print(loadings)
```

Table2: Key construction Code snippet of PCA + KMeans Clustering

PC1 (29.3% variance) loads positively on GDP and Reserves , it represents economic scale. PC2 (24.7%) loads on trade openness and inflation ,it represents external balance and monetary stability. StandardScaler is deliberately used here (not Min-Max) because PCA is sensitive to variable scale; without it, GDP and Reserves would dominate all components simply by magnitude. Four distinct structural regimes emerge: dominant global powers (top-right), stable open economies, emerging markets, and commodity-dependent/high-inflation economies.

4.6 Dynamic Repositioning Scatter — 2008 vs 2020

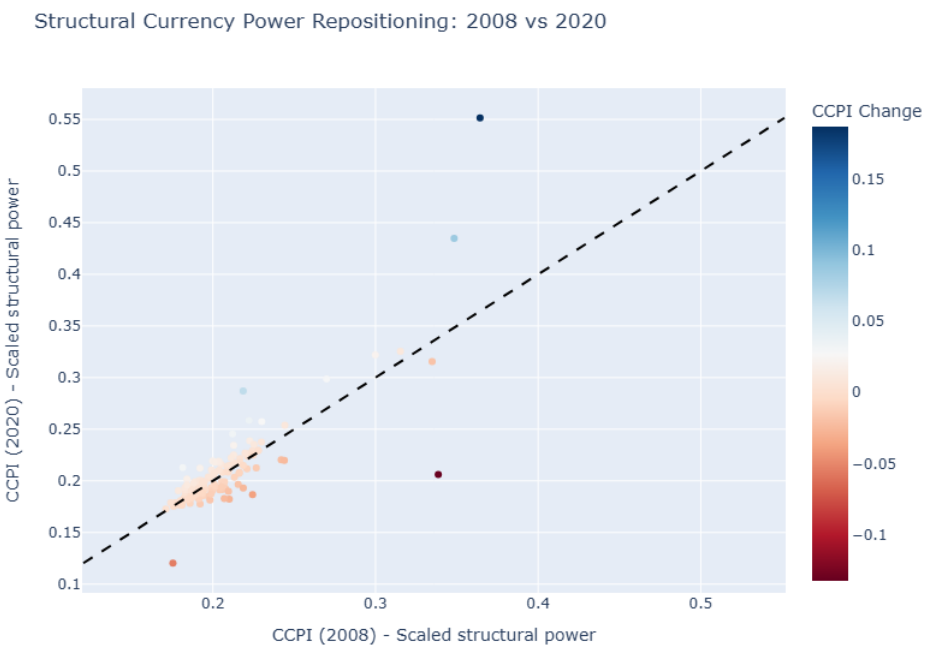

Figure 6: Dynamic Repositioning Scatter — CCPI 2008 vs 2020 (above line = gained power)

```
data = structural_scatter_dynamic(year1, year2)

# Create the advanced scatter plot
import plotly.express as px

fig = px.scatter(
    data,
    x=year1,
    y=year2,
    color="Change",
    color_continuous_scale="RdBu",
    hover_name="Country",
    size_max=15,
    title=f"Structural Currency Power Repositioning: {year1} vs {year2}"
)

# Dynamic 45-degree reference line
min_val = min(data[year1].min(), data[year2].min())
max_val = max(data[year1].max(), data[year2].max())

fig.add_shape(
    type="line",
    x0=min_val, y0=min_val,
    x1=max_val, y1=max_val,
    line=dict(dash="dash", color="black", width=2)
)
```

Table 3: Key Code snippet of *Dynamic Repositioning Scatter*

Countries above the 45° dashed line gained structural currency power between 2008 and 2020. China is the largest gainer (+0.187), followed by the United States (+0.086) and Switzerland (+0.069). Ireland's appearance among the top five gainers (+0.036) reflects its transformation into a hub for multinational profits and financial services. Losers include Timor-Leste (−0.133), Sudan, Libya, and Algeria , all hydrocarbon-dependent economies hit by commodity revenue collapse and reserve drawdowns.

4.7 Animated Bar Chart Race

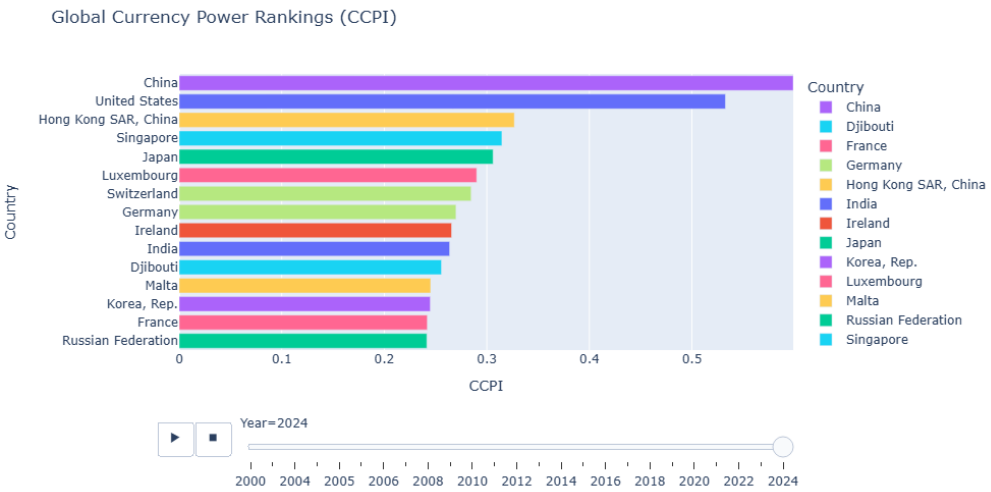

Figure 7: Animated Bar Chart Race — Global Currency Power Rankings (CCPI), 2000–2024

China enters the top ten around 2003 and claims first position by the mid-2000s, where it remains through 2024. The United States, Germany, and Japan are consistent fixtures. Small financial centres like Singapore, Hong Kong, Bahrain consistently feature due to their exceptional trade openness and reserve-to-GDP ratios, a structural pattern that globalisation reinforces rather than diminishes.

5. Theoretical and Design Considerations

5.1 Colour Scale Selection: Plasma (choropleth) was chosen for perceptual uniformity and colour-vision accessibility. Coolwarm (heatmap) is appropriate where zero has a meaningful centre. RdBu (repositioning scatter) uses red for loss and blue for gain ,consistent with financial conventions.

5.2 Animation vs Static: Animation is used only where temporal change is the primary question (choropleth, bubble chart, bar race). The repositioning scatter resolves animation's cognitive load by making temporal comparison the explicit structure , two years become two axes, enabling direct simultaneous comparison.

5.3 Weighting and Normalisation: GDP carries the highest weight (30%) as the primary driver of systemic economic scale. Reserves (25%) represent direct capacity to defend a currency. Min-Max normalisation produces bounded [0,1] scores directly readable as a proportion of maximum observed power, preserving the full variation across 100+ countries rather than compressing it around outliers.

6. Aspects of Data Analysis Investigated

- **Pattern Recognition:** PCA + K-Means identifies four latent structural economic regimes not visible from individual variable plots.
- **Distribution Comparison:** Choropleth map compares CCPI distributions across geographic regions and across time.
- **Composite Scoring & Ranking:** CCPI tracks country rankings annually over 25 years; bar race makes these dynamics accessible.
- **Correlation Analysis:** Heatmap confirms that all six components contribute independent information to the index.
- **Temporal & Shock Analysis:** Repositioning scatter and trajectory lines identify structural winners and losers during the 2008 GFC and 2020 COVID shock.

7. Novel Aspects of This Approach

The CCPI itself is the primary novel contribution: no published work combines these six WDI variables into a single normalised composite and tracks it across 100+ countries over 25 years. The dynamic repositioning scatter function is parameterised to accept any two user-specified years, making it a reusable analytical tool not found in existing WDI visualisations. Applying the animated bar race to a composite structural power index rather than raw GDP provides a more nuanced picture of global economic hierarchy than existing bar race charts.

8. Complex Implementation Detail

The PCA + K-Means pipeline (Section 4.5) is the most technically demanding implementation. Its complexity lies in the sequential application of four operations filtering, StandardScaler normalisation, K-Means clustering, and PCA projection each requiring a distinct methodological decision: which year to filter to, which scaler to use (StandardScaler for PCA, not Min-Max), how to choose K (elbow curve), and how to interpret the PCA loadings back into economic meaning. The variance explained annotations (29.3%, 24.7%) and cluster colour coding are embedded to make the chart self-documenting. This process demonstrates how unsupervised machine learning and dimensionality reduction can reveal structural typologies that descriptive charts alone cannot surface.

9. Limitations and Future Improvements

The CCPI weighting scheme is theoretically motivated rather than empirically derived; regression-based or PCA-derived weights would be more data-driven. Forward-fill imputation may introduce bias for countries with extended data gaps, disproportionately affecting lower-income nations. The exchange rate variable was excluded because its directional effect on currency power is ambiguous without conditioning.

Given more time: (1) a Dash/Shiny web app allowing users to adjust weights and compare custom periods; (2) sensitivity analysis via Monte Carlo simulation on the weighting scheme; (3) hierarchical clustering with silhouette validation to complement K-Means; (4) integration of a secondary dataset (e.g. World Governance Indicators) to explore whether structural currency power correlates with governance quality.

10. Data Ethics

All data is sourced from World Bank Open Data under an open access licence. No personally identifiable information is used. The CCPI is constructed for academic purposes only and should not be used for financial or policy decisions. Scores for countries with above-average data missingness should be interpreted with caution.

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